Lecture-30
External Flows III: Momentum Integral Equation

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Learning Objectives

By the end of this lecture, you should be able to

1. Differentiate between **differential analysis** and **control volume integral methods** for boundary layers
2. Derive the **momentum integral equation** using mass and momentum conservation principles
3. Compute **boundary layer thickness** and **wall shear stress** by applying **assumed velocity profiles**

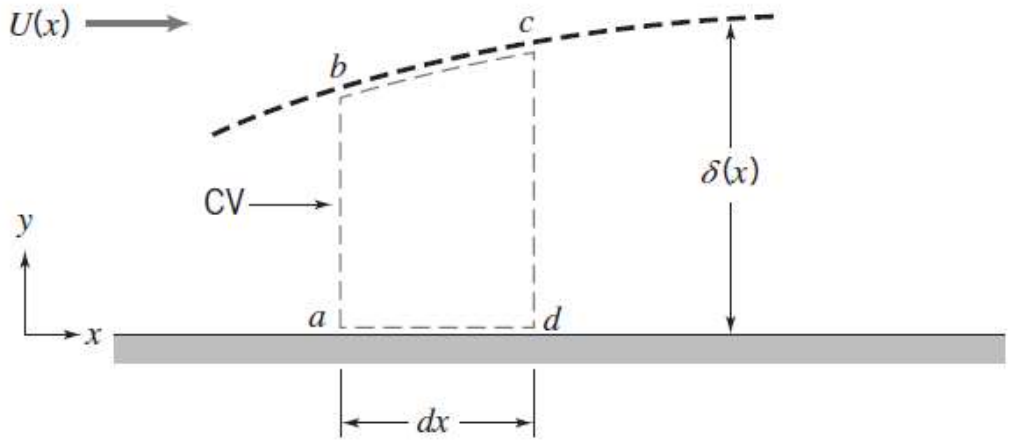
Need for a momentum integral approach

- The exact solution for laminar boundary layers over a flat plate is **mathematically complex**.
- We need a more general theory that can help us analyze **both laminar and turbulent** boundary layers.
- Instead of using a differential approach, we will apply the **conservation laws for a control volume**.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Problem setup and basic results

- We assume that the flow is **2D**, **steady** and **incompressible**.
- First, we **approximate the free-stream velocity** using the governing equations for inviscid flow.
- **Assume a velocity profile** for the flow inside the boundary layer.
- Use the momentum integral equation to calculate quantities of interest.



$$-\delta\frac{dp}{dx}-\tau_w=\frac{\partial}{\partial x}\int_0^{\delta}u\,\rho u\,dy-U\frac{\partial}{\partial x}\int_0^{\delta}\rho u\,dy$$

$$\frac{\tau_w}{\rho}=\frac{d}{dx}(U^2\theta)+\delta^* U\frac{dU}{dx}$$

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Revisiting the learning objectives

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